

Research Notes on Mean squared error

Mean squared error

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

>> Section 1

$$S_a^2 = \frac{n-1}{n} S_{n-1}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

$$\{ \displaystyle S_a^2 = \frac{n-1}{n} S_{n-1}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \}$$

>> Section 2

$$\text{MSE} = (S_{n-1}^2) = \frac{1}{n} (\mu^2 - \frac{1}{n} \sum_{i=1}^n x_i^2) = \frac{1}{n} (\gamma^2 + \frac{2}{n} \sum_{i=1}^n x_i - \frac{1}{n} \sum_{i=1}^n x_i^2)$$

$$\{ \displaystyle \text{MSE} = (S_{n-1}^2) = \frac{1}{n} (\mu^2 - \frac{1}{n} \sum_{i=1}^n x_i^2) = \frac{1}{n} (\gamma^2 + \frac{2}{n} \sum_{i=1}^n x_i - \frac{1}{n} \sum_{i=1}^n x_i^2) \}$$

>> Section 3

An even shorter proof can be achieved using the well-known formula that for a random variable X , $E(X^2) = \text{Var}(X) + (E(X))^2$. By substituting X with $\theta - \hat{\theta}$, we have

>> Section 4

$$\text{MSE} = (S_a^2) = E[(\frac{n-1}{n} S_{n-1}^2 - \sigma^2)^2] = E[(\frac{n-1}{n})^2 S_{n-1}^4 - 2(\frac{n-1}{n}) S_{n-1}^2 \sigma^2 + \sigma^4]$$

$$= (\frac{n-1}{n})^2 E[S_{n-1}^4] - 2(\frac{n-1}{n}) E[S_{n-1}^2] \sigma^2 + \sigma^4 = (\frac{n-1}{n})^2 E[S_{n-1}^4] - 2(\frac{n-1}{n}) \sigma^2 + \sigma^4$$

$$= (\frac{n-1}{n})^2 E[S_{n-1}^4] - 2(\frac{n-1}{n}) \sigma^2 + \sigma^4 = (\frac{n-1}{n})^2 E[S_{n-1}^4] - 2(\frac{n-1}{n}) \sigma^2 + \sigma^4$$

$$= (\frac{n-1}{n})^2 E[S_{n-1}^4] - 2(\frac{n-1}{n}) \sigma^2 + \sigma^4 = (\frac{n-1}{n})^2 E[S_{n-1}^4] - 2(\frac{n-1}{n}) \sigma^2 + \sigma^4$$

$$\{ \displaystyle \text{MSE} = (S_a^2) = E[(\frac{n-1}{n} S_{n-1}^2 - \sigma^2)^2] = E[(\frac{n-1}{n})^2 S_{n-1}^4 - 2(\frac{n-1}{n}) S_{n-1}^2 \sigma^2 + \sigma^4] \}$$

$$(S_{n-1}^2) = \frac{1}{n} (\mu^2 - \frac{1}{n} \sum_{i=1}^n x_i^2) = \frac{1}{n} (\gamma^2 + \frac{2}{n} \sum_{i=1}^n x_i - \frac{1}{n} \sum_{i=1}^n x_i^2)$$

$$\{ \displaystyle (S_{n-1}^2) = \frac{1}{n} (\mu^2 - \frac{1}{n} \sum_{i=1}^n x_i^2) = \frac{1}{n} (\gamma^2 + \frac{2}{n} \sum_{i=1}^n x_i - \frac{1}{n} \sum_{i=1}^n x_i^2) \}$$

>> Section 5

In statistics, the mean squared error (MSE) or mean squared deviation (MSD) of an estimator (of a procedure for estimating an unobserved quantity) measures the average of the squares of the errors—that is, the average squared difference between the estimated values and the true value. MSE is a risk function, corresponding to the expected value of the squared error loss. The fact that MSE is almost always strictly positive (and not zero) is because of randomness or because the estimator does not account for information that could produce a more accurate estimate. In machine learning, specifically empirical risk minimization, MSE may refer to the empirical risk (the average loss on an observed data set), as an estimate of the true MSE (the true risk: the average loss on the actual population distribution). The MSE is a measure of the quality of an estimator. As it is derived from the square of Euclidean distance, it is always a positive value that decreases as the error approaches zero. The MSE is the second moment (about the origin) of the error, and thus incorporates both the variance of the estimator (how widely spread the estimates are from one data sample to another) and its bias (how far off the average estimated value is from the true value). For an unbiased estimator, the MSE is the variance of the estimator. Like the variance, MSE has the same units of measurement as the square of the quantity being estimated. In an analogy to standard deviation, taking the square root of MSE yields the root-mean-square error or root-mean-square deviation (RMSE or RMSD), which has the same units as the quantity being estimated; for an unbiased estimator, the RMSE is the square root of the variance, known as the standard error.

>> Section 6

$$a = (\frac{n-1}{n}) \gamma^2 + \frac{2}{n} \sum_{i=1}^n x_i + \frac{1}{n} \sum_{i=1}^n x_i^2$$

$$\{ \displaystyle a = (\frac{n-1}{n}) \gamma^2 + \frac{2}{n} \sum_{i=1}^n x_i + \frac{1}{n} \sum_{i=1}^n x_i^2 \}$$

>> Section 7

Predictor If a vector of n predictions is generated from a sample of n data points on all variables, and Y is the vector of observed values of the variable being predicted, with $\hat{Y}$ being the predicted values (e.g. as from a least-squares fit), then the within-sample MSE of the predictor is computed as

Research Notes on Mean squared error

Mean squared error

>> Section 8

$$\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] = \text{Var}(\hat{\theta}) + (E[\hat{\theta}] - \theta)^2 = \text{Var}(\hat{\theta}) + \text{Bias}^2(\hat{\theta}, \theta)$$

>> Section 9

where e_i is $Y_i - \hat{Y}_i$ and $\mathbf{e}$ is a $n \times 1$ column vector. The MSE can also be computed on q data points that were not used in estimating the model, either because they were held back for this purpose, or because these data have been newly obtained. Within this process, known as cross-validation, the MSE is often called the test MSE, and is computed as